

Tele-Leprosy Triage Console — Workflow Document

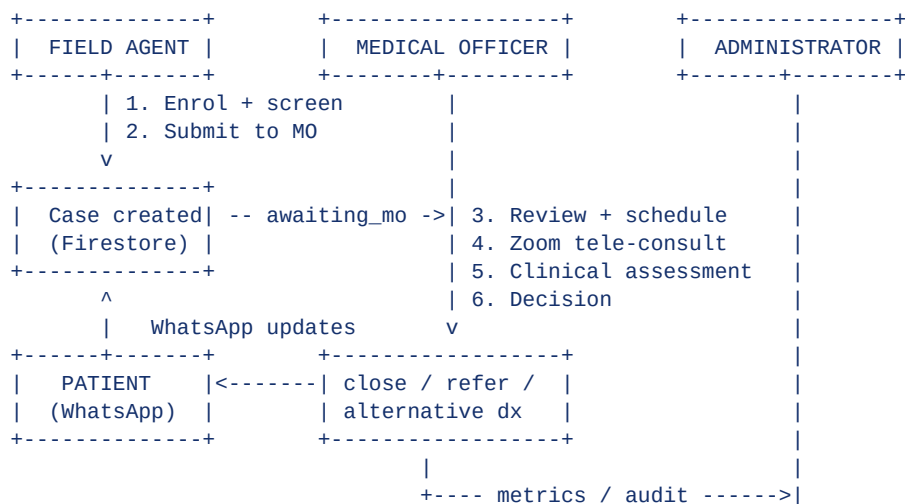
A field-to-specialist tele-leprosy screening and triage platform for the SHAKTHI programme (Bastar district, Chhattisgarh). Field agents screen patients on-site, data is sent to a Medical Officer (MO) for tele-consultation and decision, and administrators oversee the operation.

1. Roles

Role	Who	What they do
Field Agent	Community health worker / mediator	Enrol patients, capture history + the 11-symptom leprosy screening on-site, submit to the MO. No clinical decision.
Medical Officer (MO)	Doctor	Review the agent's submission, run a Zoom tele-consult, record a clinical assessment, and issue a decision (close / alternative diagnosis / refer).
Administrator	Programme admin	Monitor the dashboard (metrics, escalations), manage users, and review the audit log. Provisioned out-of-band (no self-registration).

Patients do **not** log in. They are enrolled by an agent and receive updates via WhatsApp.

2. High-level flow



3. Authentication & onboarding

- **Login page** has three role cards: **Field Agent**, **Medical Officer**, **Administrator**.
 - A card only logs in its own role — signing in with another role's credentials is rejected and signed out.
- **Agent & MO** can self-register (Sign in / Register toggle). Registration captures **name**, **phone (10 digits)**, **email**, **password**. Phone can be changed later in **Settings -> Profile**.

- **Administrator** is **login-only** (provisioned by an admin; no public sign-up).
- Auth is **Firestore Authentication**; the role lives on the user's Firestore profile and is mirrored to a custom token claim.
- Phone numbers are used to send WhatsApp notifications to agents and MOs.

4. Field Agent workflow (3-step intake wizard)

The agent intake is a single wizard that batches all writes for **offline safety** (works with no connectivity; syncs when back online).

Step 1 — Enrollment

- **Programme context:** select **PHC / CHC** (Bakawand, Karpawand, Kolawal, Mangnaar, Kachnaar, Maalgaon, Jebel). Selecting a PHC **auto-fills State (Chhattisgarh) and District (Bastar)**.
- **Patient identity:** name, age, sex, **phone (exactly 10 digits)**.
- **Contact & location:** state/district (auto), village, gram panchayat, house no.
- **Household:** household number, relation to head, head-of-family name + phone.
- **Health identifiers:** Aadhaar (12 digits), ABHA (14 digits) — optional.
- Consent checkbox (required).

Step 2 — History

- **Chronic conditions:** Diabetes, Hypertension, TB, HIV, Pregnancy, None.
 - "Pregnancy" is hidden for male patients.
- Optional prior prescription / lab photos and past-visit notes.

Step 3 — Symptoms

- **Screening context:** Date of screening **auto-set to today**; **GPS location auto-captured** on entry (re-capturable).
- **11-question leprosy symptom checklist** (numbered, Yes/No each):
 1. Light-coloured or reddish skin patch(es)
 2. Reduced or loss of sensation over skin patch(es)
 3. Tingling, numbness, or burning sensation in hands/feet
 4. Weakness in hands or feet
 5. Weak grip or objects slipping from hands
 6. Painless wounds, burns, or ulcers on hands/feet
 7. Pain or tenderness near elbow, wrist, knee, or ankle
 8. Foot slipping out of slippers/chappals or dragging while walking
 9. Difficulty closing eyes completely or reduced blinking
 10. Loss of eyebrows, collapsed nose
 11. Lumps/nodules on skin or swelling of earlobes
- **Lesion photos** and **lab report** uploads.

- **Submit to Medical Officer** — the agent makes **no decision**; every case is routed to the MO queue.

What happens on submit

1. The full intake (patient + history + screening + images) is queued in IndexedDB and uploaded (immediately if online, on reconnect if offline).
2. Backend creates the **patient + case**, stores the screening, and computes a **leprosy risk summary** (High / Moderate / Low) for the MO's reference only.
3. Case status -> **awaiting_mo**.
4. Artefacts are gathered into the patient's Storage folder (agent report PDF, images, labs — see §8).

5. Medical Officer workflow

Queue

- The MO sees their cases (patient name, case ID, status). Risk labels are intentionally **not** shown here.

Case Review (collapsible cards to reduce scrolling)

1. **Tele-consult** — schedule a Zoom slot (date/time + duration). Booking:
 - Creates a Zoom meeting and sends the **join link via WhatsApp to the patient, the agent, and the MO**.
 - Case status -> **scheduled**.
 - **"Start Zoom consult"** is enabled around the slot time (host link opens Zoom).
 - **Screen recording**: the MO can record the consultation (browser screen capture); the recording uploads to the patient folder.
2. **Intake by field agent** (collapsible) — full read-only view of demographics, history, the 11 symptom answers, lesion images, and lab reports.
3. **MO clinical assessment** (collapsible, numbered 1–9): confirmed leprosy, no. of skin lesions, nerve involvement grid, status of case, WHO classification, sensory loss, disability grade, complications, treatment plan. **Must be saved before a decision.**
4. **Post-consultation** — upload reports/prescriptions (PDF/image) and view recordings; all stored in the patient folder.
5. **Decision** — one of:
 - **Close at community level** (`close_remote`)
 - **Alternative diagnosis** (`alt_dx`)
 - **Refer** (`refer`)

On decision ("Save decision & notify")

- The decision is **saved first** (independent of WhatsApp), the **MO report PDF** is generated and stored in the patient folder.
- A WhatsApp message (with the MO report attached) is sent to the **patient and the agent** — **not** the MO.
- Case status -> `closed_remote` / `closed_alt_dx` / `referred`.

Expired sessions = read-only

- If a consult slot has **passed**, the case becomes **read-only**: a banner is shown, the clinical assessment fields are disabled, and the decision cannot be changed. Everything remains viewable/downloadable.

6. Administrator workflow

- **Dashboard** — operational metrics (case counts by outcome, recent escalations, pending image cases). "View all patients" opens the patient directory.
- **Users** — list/manage staff accounts and roles (agent / mo / admin).
- **Audit** — system audit log.

7. Notifications (WhatsApp)

Event	Recipients	Content
Tele-consult scheduled	Patient + Agent + MO	Zoom join link, time, duration
MO decision issued	Patient + Agent (not MO)	Outcome + next step + MO report PDF
Community close / rule-out (legacy)	Patient	Outcome + advice

Requires approved Meta WhatsApp templates: [appointment_scheduled](#), [triage_decision](#), [triage_decision_with_report](#). Recipients without a saved phone are skipped gracefully.

8. Document storage (per-patient folder)

All artefacts for a patient are organised under one Storage folder:

```
patients/&lt;case-id&gt;__&lt;patient-name-slug&gt;/
+-- agent-report-&lt;id&gt;.pdf          # generated on screening submit
+-- mo-report-&lt;id&gt;.pdf            # generated on MO decision
+-- images/image-1.jpg ...          # lesion photos from intake
+-- labs/lab-1.jpg ...              # lab reports from intake
+-- consultation/&lt;ts&gt;-&lt;file&gt;      # MO post-consult uploads
+-- recordings/consultation-&lt;ts&gt;.webm # screen recordings
```

- **Agent Report** = the intake PDF (everything the agent captured).
- **MO Report** = the decision PDF (clinical assessment + final decision).

9. Case status lifecycle

```
intake -> awaiting_mo -> scheduled -> (consult) -> closed_remote
                                                +> closed_alt_dx
                                                +> referred
```

([closed_rule_out](#) is also used for rule-out closes.)

10. Architecture & tech stack

Layer	Technology
Frontend	React + Vite + Tailwind CSS, PWA (installable, offline-first via IndexedDB + service worker)
Backend	FastAPI (Python)
Auth	Firebase Authentication (email/password) + role custom claims
Database	Cloud Firestore (users , patients , cases , appointments , notifications , recalls)
File storage	Firebase Storage (per-patient folders)
Video	Zoom (meeting creation + Meeting SDK signature; host/participant links)
Messaging	WhatsApp Business (Meta) templates
PDF	ReportLab (agent report + MO report)

Offline-first

The agent intake is fully usable offline: the bundle (patient + history + screening + image blobs) is stored in IndexedDB and uploaded by the sync engine when connectivity returns. Forced submission always routes to the MO.

11. Operational prerequisites (deployment checklist)

1. **Firebase project** with Authentication, Firestore, and **Storage enabled** (the default bucket must exist — required for report/recording uploads).
2. **WhatsApp Business templates** created and approved in Meta: [appointment_scheduled](#), [triage_decision](#), [triage_decision_with_report](#).
3. **Zoom** API credentials (for meeting creation) and Meeting SDK key/secret.
4. Service-account credentials configured for the backend ([FIREBASE_CREDENTIALS_*](#), [FIREBASE_STORAGE_BUCKET](#)).
5. At least one **admin** provisioned out-of-band; agents and MOs self-register and add their phone.

12. End-to-end summary (one patient)

1. Agent opens the app (online or offline), enrolls the patient, records history and the 11-symptom screening with photos, and **submits to the MO**.
2. The case appears in the MO queue ([awaiting_mo](#)); the agent report PDF + images are stored in the patient folder.
3. The MO **schedules a Zoom consult** -> patient, agent, and MO get the link by WhatsApp.
4. The MO runs the consult (optionally **recording the screen**), fills the **clinical assessment**, and issues a **decision**.
5. The **MO report PDF** is generated and the decision is sent by WhatsApp to the **patient and agent**; the case is closed/referred.

6. After the slot expires, the case is **read-only**.
7. Admins monitor the whole operation via the **dashboard** and **audit log**.